

GIRIJANANDA CHOWDHURY UNIVERSITY, AZARA, GUWAHATI 781017, ASSAM
DEPARTMENT OF CIVIL ENGINEERING
Formative Assessment -1

Course: BCE23404T (Hydraulic Engineering)
Date: 25/02/2025
Duration: 1 hour

Academic Year: 2024-25
Total Marks: 30

Question		Marks	Course Outcome (s)
1	Derive an expression for the velocity distribution for viscous flow through a pipe. Also sketch the velocity distribution and shear stress distribution across a section of the pipe.	15	CO2
2	Calculate : (i) the pressure gradient along flow, (ii) the average velocity and (iii) the discharge for an oil of viscosity 0.02 Ns/m^2 flowing between two stationary parallel plates 1m wide maintained 10mm apart. The velocity midway between the plate is 2m/s .	10	CO2
3	Derive an expression for loss of head due to friction in pipes.	5	CO1